

Photography problems — Solution

X26 (2026) — English translation

A1

2.00 pts

Find the position and shape of the wire if its length is minimal.

The image of the wire is symmetrical about Y , so the wire lies in the xy plane. Note that each point of the wire gives a point on the screen (if the camera is focused on this point) or a circle (from the likeness of cones).

Let's find the position of the circle and its dimensions for the point with coordinates $(x, y, 0)$. Coordinates of the center of the circle from the similarity of triangles:

$$(Y, Z) = \left(\frac{yl}{x}, 0 \right).$$

For diameter:

$$\frac{d}{D} = \frac{a-l}{a} = 1 - \frac{l}{a}.$$

Substitute value $1/a$ from the thin lens formula.

$$\frac{d}{D} = 1 - l \left(\frac{1}{f} - \frac{1}{x} \right) = 1 - \frac{l(x-f)}{fx}.$$

Find dependence d from Y for $Y > 0$.

By condition $\tan \theta = k$.

$$d = 2Y \sin \theta = \frac{2k}{\sqrt{1+k^2}} \cdot Y = \gamma Y = \gamma \cdot \frac{yl}{x},$$

where $\gamma = \frac{2k}{\sqrt{1+k^2}}$. Substituting this into the previously obtained expression for d .

$$\frac{d}{D} = \frac{\gamma yl}{Dx} = 1 - \frac{l(x-f)}{fx}$$

Express $y(x)$:

$$y = \frac{Dx}{\gamma l} - \frac{D(x-f)}{\gamma f} = \frac{D}{\gamma} - \frac{Dx}{\gamma} \left(\frac{1}{f} - \frac{1}{l} \right) = \frac{D}{\gamma} - \frac{Dx(l-f)}{\gamma fl}.$$

Note, that since wire place in region $x > 0$ and image point wire in center screen be point, that camera focus on one from point wire, therefore $l > f$. Note, that we consider case $a \geq l$, if this not thus, that the same equation remains valid, but the sign of d flips. That is, for the consideration $a < l$ sufficiently in obtain expression change sign γ . Thus we obtain, that possible position point determine form image:

$$y = \pm \frac{D(l-f)\sqrt{1+k^2}}{2klf} \left(\frac{lf}{l-f} - x \right).$$

Perhaps the wire contains more sections, the images of which overlap with the images of points on these lines, but then the wire will have a greater length, which contradicts the condition. Therefore, it lies completely on these straight lines.

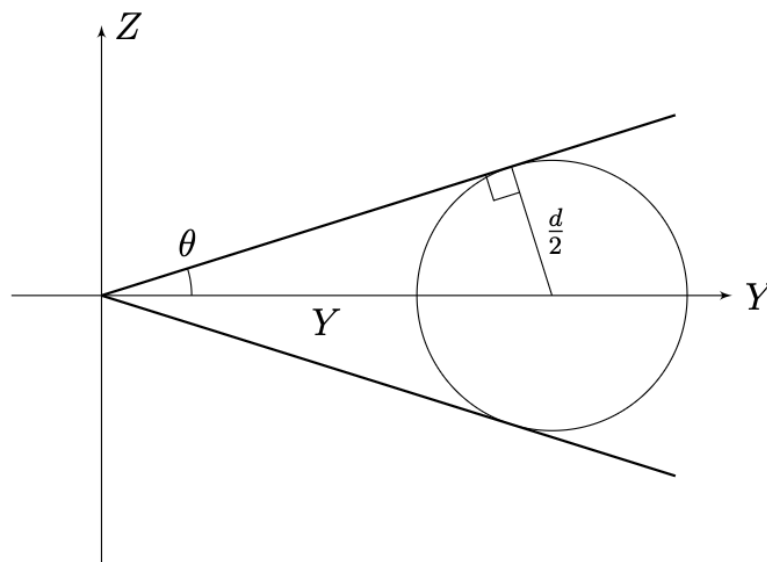
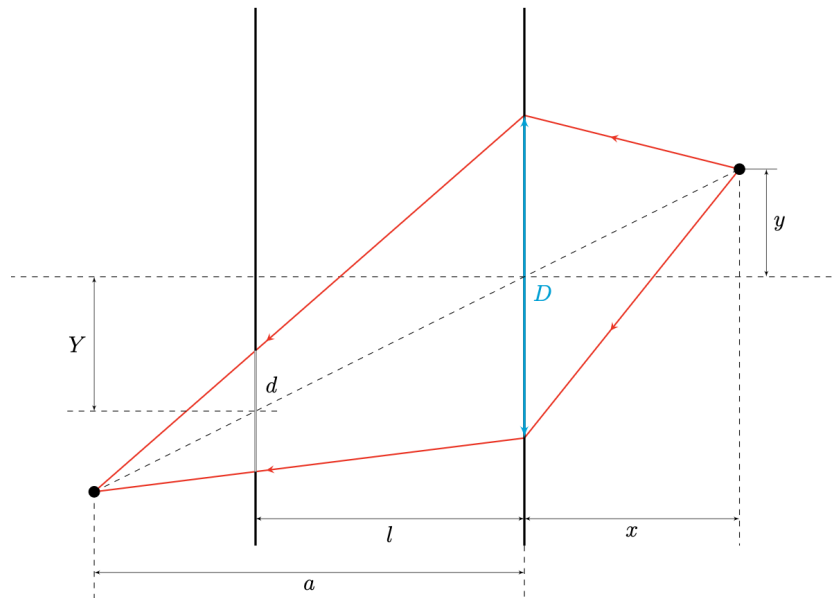
Note that the angular size of the parts of straight lines $1'$ and $2'$ is less than 90° , therefore they can give in the image a circle with a limited segment of the positions of the centers; even for a fixed angular

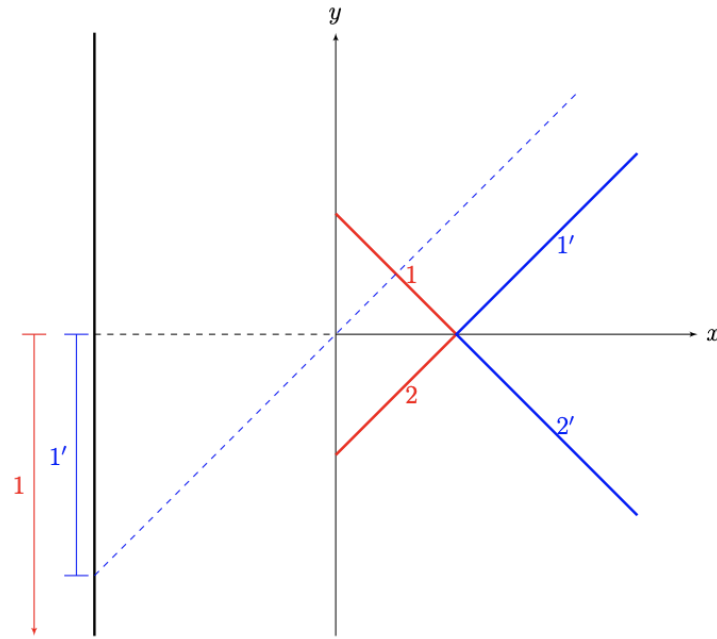
size, the length of the wire in sections 1 and 2 will always be less, therefore the wire has the shape of an angle from the segments 1 and 2:

Answer:

$$y = \pm \frac{D(l-f)\sqrt{1+k^2}}{2klf} \left(\frac{lf}{l-f} - x \right)$$

for $x \in \left[0, \frac{lf}{l-f} \right]$.





B1

2.00 pts

Using a compass and straight edge, draw a straight line parallel to the direction of the puck's speed.

Let's understand how an image is formed. Let's enter the coordinates XY , which are attached to the puck and indicate the position of the point on it, and xy , which determine the position in the photograph. Let the speed of the line that the camera films at a particular moment is equal to u ; the center of the puck was photographed at point (x_0, y_0) and at time t_0 .

The angle between the speed of the puck and the x axis will be denoted by α . Let's find the coordinates in the photograph for an arbitrary point on the puck (X, Y) . The moment of photographing this point:

$$t = t_0 + \frac{Y}{u - v \sin \alpha}.$$

Coordinate equal:

$$x = x_0 + X + v \cos \alpha (t - t_0) = x_0 + X + \frac{v \cos \alpha}{u - v \sin \alpha} Y,$$

$$y = y_0 + u (t - t_0) = y_0 + \frac{u}{u - v \sin \alpha} Y.$$

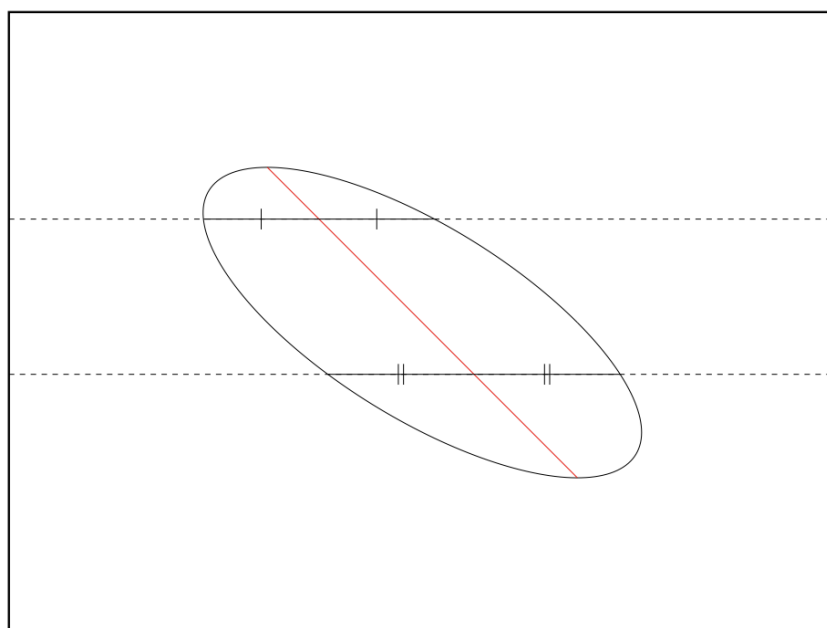
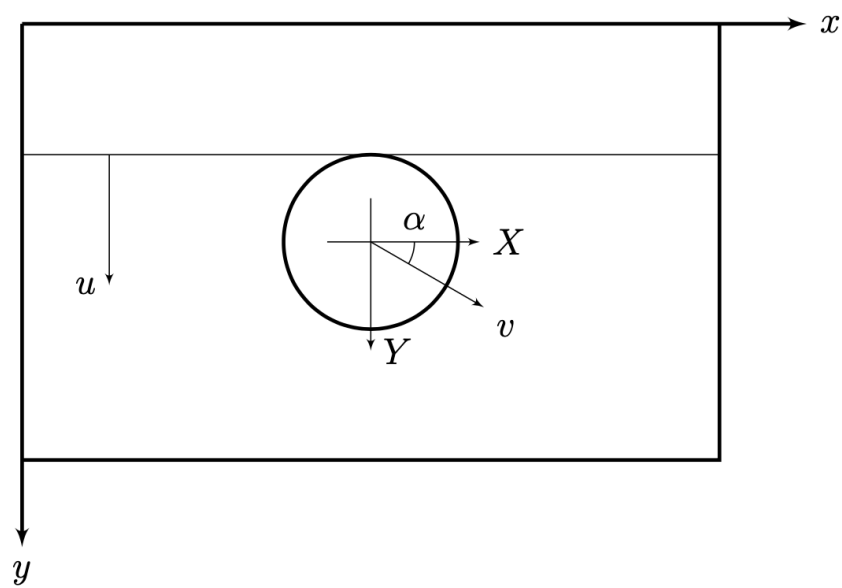
Note, that transformation coordinate linearly. Means, straight on puck be straight on image. Construct possible direction velocity. Obtain image straight $X = 0$, for this conduct horizontal straight, image $X = 0$ be straight pass through center intersection horizontal straight with image puck (since for record Y and correspondingly y coordinate x and X differ only by a constant).

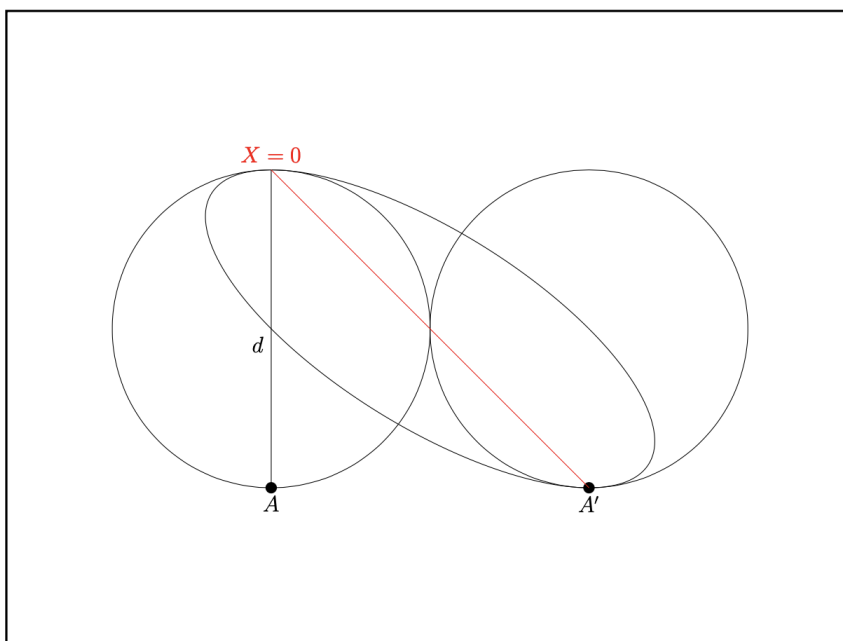
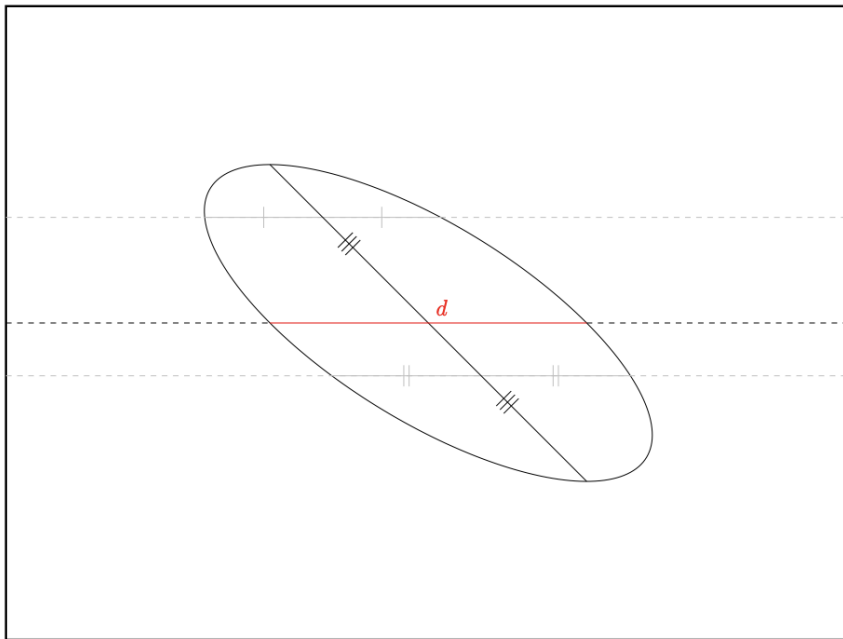
Now we get a segment of length equal to the diameter. To do this, it is enough to draw a horizontal segment through the center of the image $X = 0$.

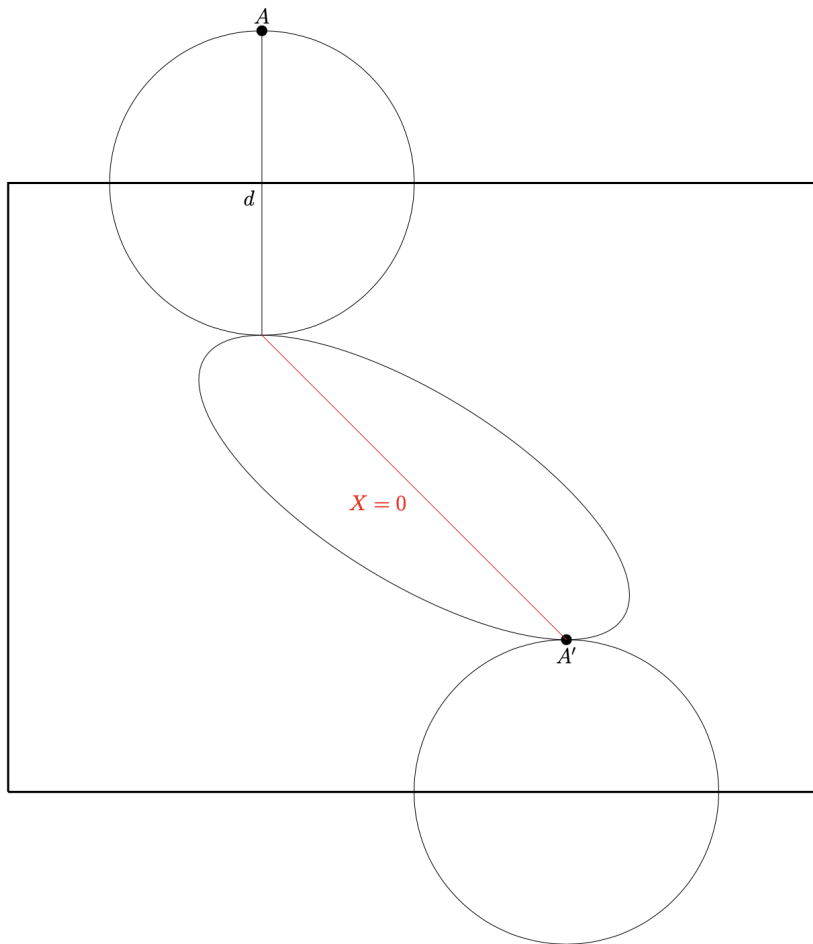
Then there are two options for how photography can take place depending on the speed of the puck. If $u > v \sin \alpha$, then the top point of the image is the top point of the washer. And accordingly, vice versa, if $u < v \sin \alpha$, then the top point of the image is the bottom point of the washer. Let's draw the initial and final states of the puck depending on the speed ratio. $u > v \sin \alpha$:

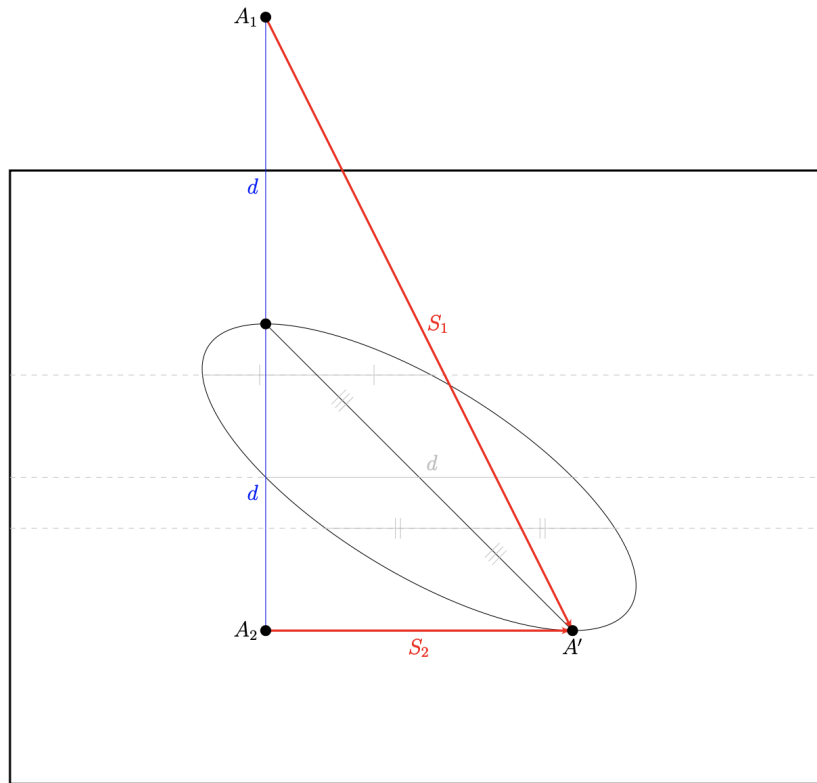
$u < v \sin \alpha$:

Note that in this way we can obtain the possible direction of the speed: we obtain the point A by constructing a vertical segment of length d with one end at the top point of the image of the puck, the speed will be directed along $\overrightarrow{AA'}$.









B2

1.00 pts

Using an additional ruler with divisions, find the speed modulus of the puck.

It remains to determine the velocity module. We plotted the displacement of point A during the time of shooting the puck τ . We can express this time in terms of the total photographing time T .

$$\tau = \frac{l}{L}T = 0.5 \cdot T = 5 \text{ ms}$$

By photograph can determine displacement puck:

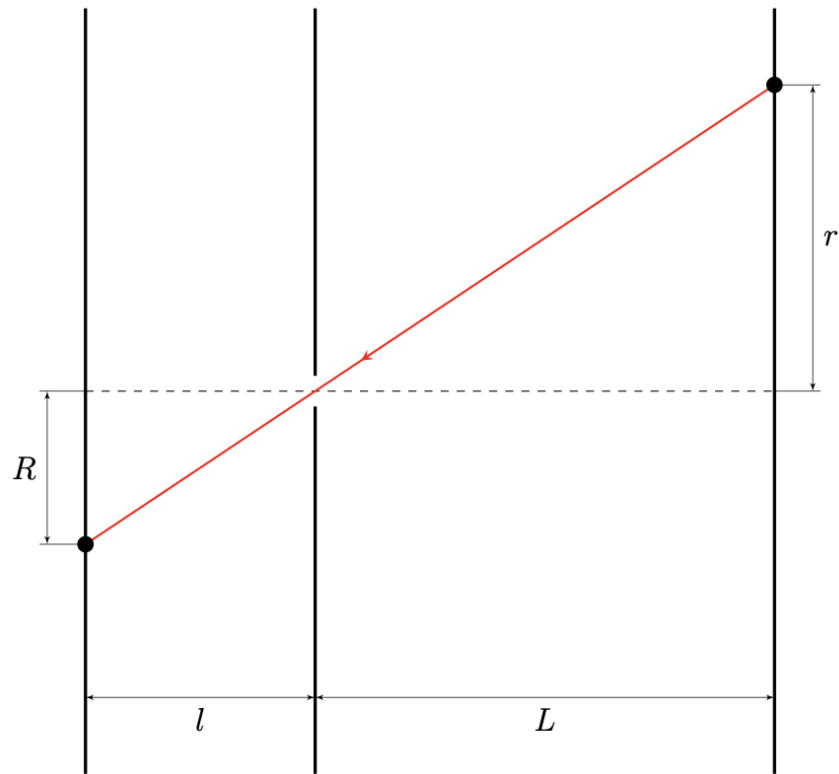
$$\frac{S_1}{d} = 2.24; \quad \frac{S_2}{d} = 1.00.$$

$$S_1 = 17.1 \text{ cm}; \quad S_2 = 7.62 \text{ cm}$$

Remain calculate quantity velocity $v = S/\tau$.

Answer:

$$v_1 = 34.1 \text{ m/s}; \quad v_2 = 15.2 \text{ m/s}$$



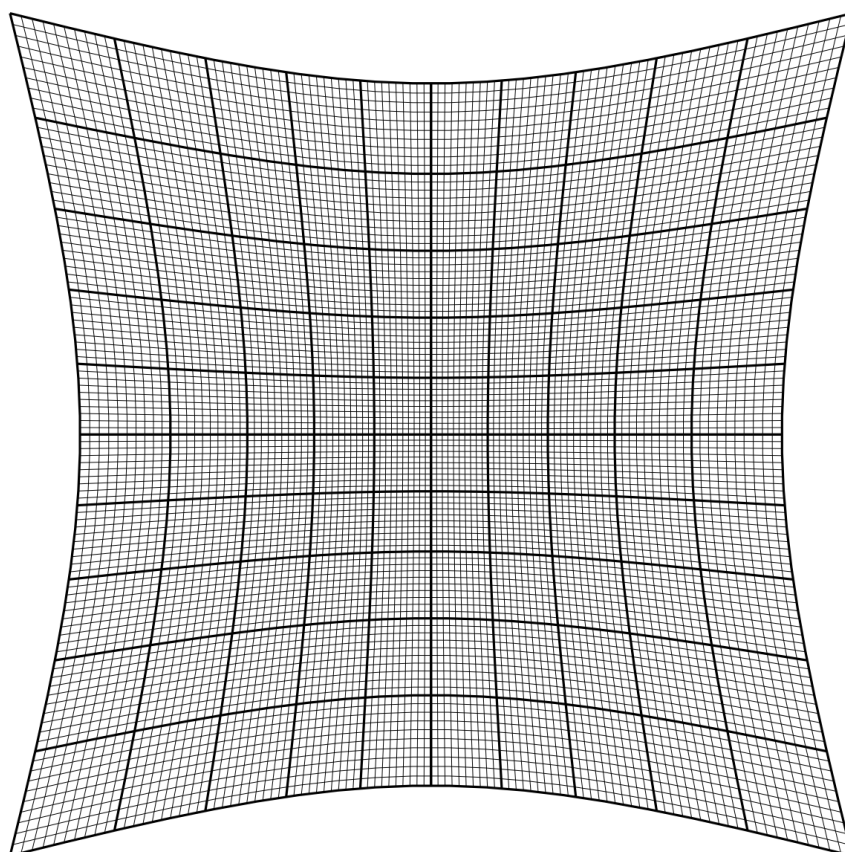
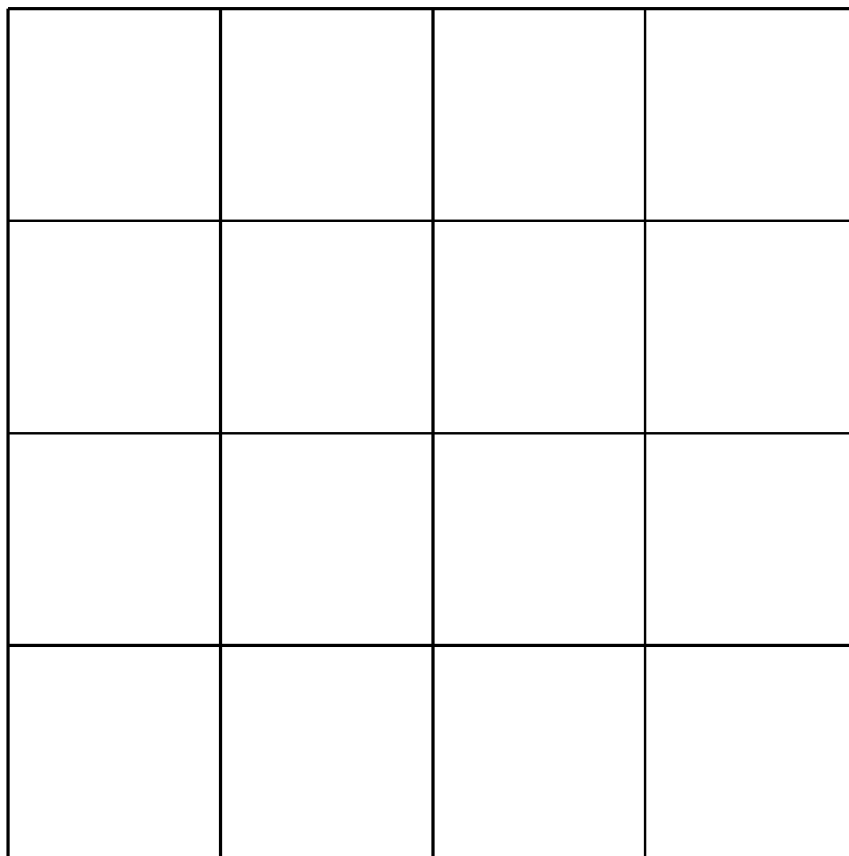
C2

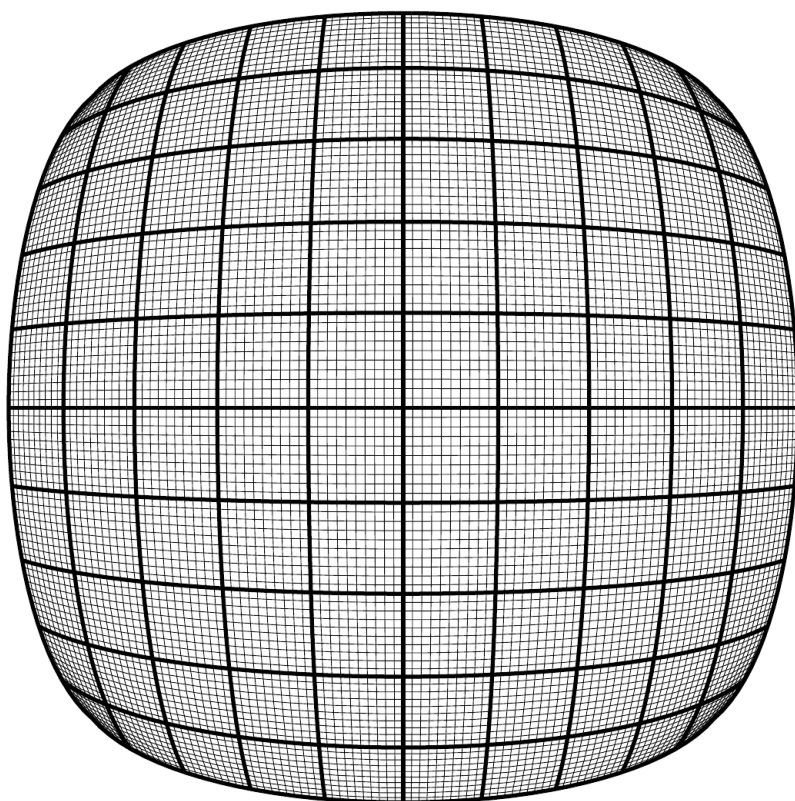
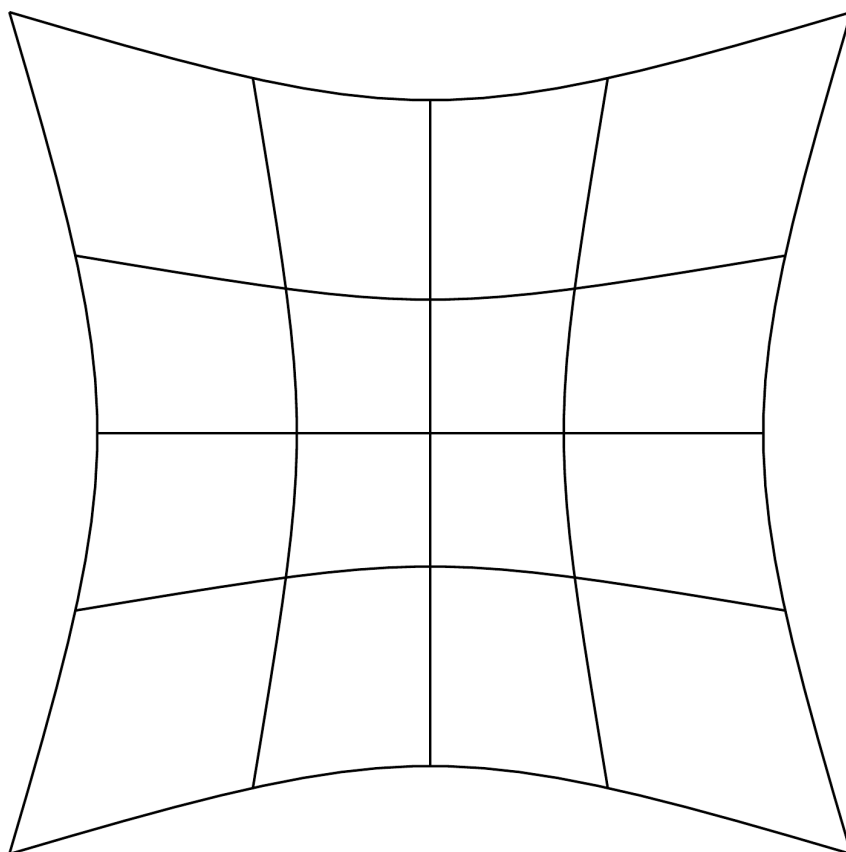
0.50 pts

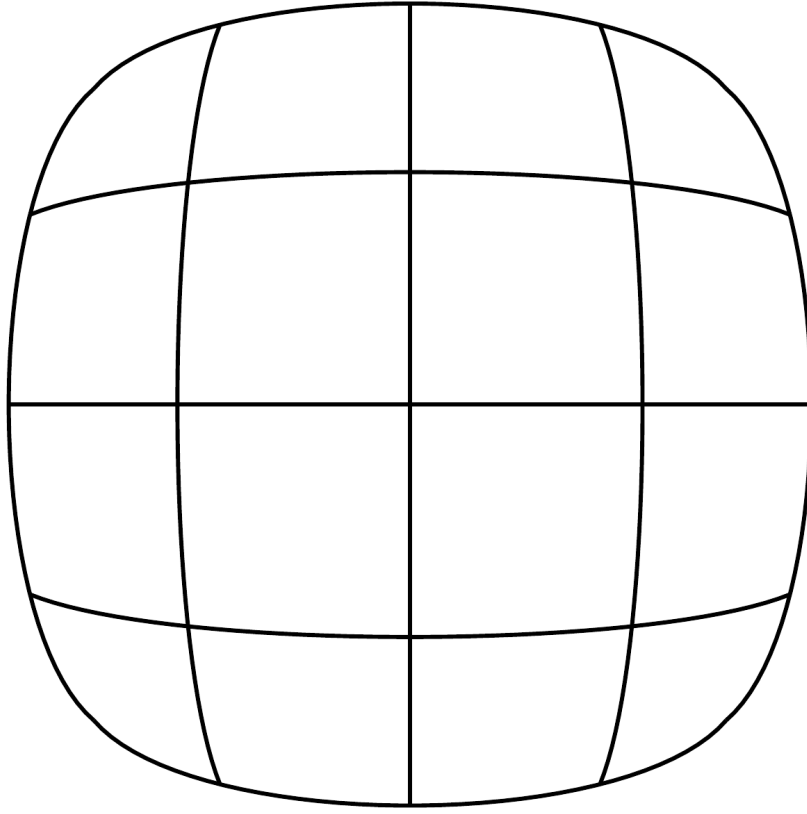
Draw a high-quality view of the photograph of part of the notebook sheet shown in the figure below, in cases $\beta > 0$ (positive distortion) and $\beta < 0$ (negative distortion). Consider that $\alpha > 0$.

In the case of $\beta > 0$ the image of the graph paper (we present it for a more detailed illustration) and the square will look like:

In case of $\beta < 0$:







C3

2.00 pts

Determine the coefficients α and β in this case. What kind of distortion is observed in such a system: positive (α and β of the same sign) or negative (α and β of different signs)?

Note that we are considering third-order effects for paraxial rays, so all formulas can be expanded to third order. Let's find the connection between R and r .

Let's express the dimensions through the angles:

$$R = l \tan \varphi;$$

$$r = (L - h) \tan \varphi + h \tan \psi.$$

Relation angle:

$$n \sin \psi = \sin \varphi.$$

$$\psi = \arcsin \left(\frac{\sin \varphi}{n} \right) \approx \arcsin \left(\frac{\varphi}{n} - \frac{\varphi^3}{6n} \right) \approx \frac{\varphi}{n} - \frac{\varphi^3}{6n} + \frac{\varphi^3}{6n^3} = \frac{\varphi}{n} - \frac{\varphi^3 (n^2 - 1)}{6n^3}$$

$$\tan \psi \approx \psi + \frac{\psi^3}{3} \approx \frac{\varphi}{n} - \frac{\varphi^3 (n^2 - 1)}{6n^3} + \frac{\varphi^3}{3n^3} = \frac{\varphi}{n} - \frac{\varphi^3 (n^2 - 3)}{6n^3}$$

$$\varphi = \arctan \left(\frac{R}{l} \right) \approx \frac{R}{l} - \frac{R^3}{3l^3}$$

$$\tan \psi \approx \frac{R}{nl} - \frac{R^3}{3l^3 n} - \frac{R^3 (n^2 - 3)}{6l^3 n^3} = \frac{R}{nl} - \frac{R^3 (n^2 - 1)}{2l^3 n^3}$$

$$r = \frac{L - h}{l} R + \frac{h}{nl} R - \frac{h (n^2 - 1)}{2l^3 n^3} R^3 = \frac{nL - (n - 1)h}{nl} R - \frac{h (n^2 - 1)}{2l^3 n^3} R^3.$$

Introduce notation:

$$A = \frac{nL - (n - 1)h}{nl}; \quad B = \frac{h (n^2 - 1)}{2l^3 n^3}.$$

$$r = AR - BR^3$$

$$r \approx A(\alpha r + \beta r^3) - B\alpha^3 r^3 = A\alpha r + (A\beta - B\alpha^3) r^3$$

Equate the coefficients at like powers and express sought coefficients:

$$\begin{cases} A\alpha = 1; \\ A\beta - B\alpha^3 = 0. \end{cases}$$

$$\alpha = \frac{1}{A} = \frac{nl}{nL - (n-1)h}$$

$$\beta = \frac{B}{A^4} = \frac{hnl(n^2 - 1)}{2(nL - (n-1)h)^4}$$

Answer: Positive distortion is observed.

$$\alpha = \frac{nl}{nL - (n-1)h} \quad \beta = \frac{hnl(n^2 - 1)}{2(nL - (n-1)h)^4}$$

